

# Apple Review Storage Schema v1

## Purpose

This schema upgrades Apple App Store review collection from temporary CSV files into a durable data model. It is designed for multi-app, multi-country, multi-source ingestion, while preserving raw review text for future sentiment analysis, topic modeling, language detection, and other NLP tasks.

The v1 model is intentionally normalized around stable entities:

- **sources**: external platforms such as Apple App Store.
- **apps**: canonical internal app records.
- **source\_apps**: source-specific app identifiers and metadata.
- **ingestion\_runs**: one execution of a collector.
- **reviews**: one review record from one source app.
- **review\_quality**: derived data-quality and deduplication signals.
- **review\_nlp\_annotations**: future NLP outputs, versioned by model/task.

## Design Principles

- Preserve raw source data and raw text.
- Keep app/source identity separate from reviews.
- Make country and language first-class fields.
- Track every ingestion run and collection timestamp.
- Support idempotent upserts by source review identity.
- Add derived quality/NLP fields without mutating raw review content.
- Keep v1 simple enough for SQLite/PostgreSQL, but compatible with larger warehouses later.

## Entity Relationship Summary

```
sources 1--many source_apps many--1 apps
source_apps 1--many reviews
ingestion_runs 1--many reviews
reviews 1--1 review_quality
reviews 1--many review_nlp_annotations
```

## Tables

### sources

Stores external data platforms.

| Column    | Type       | Required | Notes        |
|-----------|------------|----------|--------------|
| source_id | integer pk | yes      | Internal key |

| Column      | Type        | Required | Notes                                          |
|-------------|-------------|----------|------------------------------------------------|
| source_key  | text unique | yes      | Example: <code>apple_app_store</code>          |
| source_name | text        | yes      | Example: <code>Apple App Store</code>          |
| base_url    | text        | no       | Example: <code>https://itunes.apple.com</code> |
| created_at  | timestamp   | yes      | UTC                                            |
| updated_at  | timestamp   | yes      | UTC                                            |

## apps

Canonical app/product identity, independent of platform.

| Column         | Type        | Required | Notes                                         |
|----------------|-------------|----------|-----------------------------------------------|
| app_id         | integer pk  | yes      | Internal key                                  |
| app_key        | text unique | yes      | Stable slug, e.g. <code>uber_eats</code>      |
| canonical_name | text        | yes      | Example: <code>Uber Eats</code>               |
| publisher      | text        | no       | Example: <code>Uber Technologies, Inc.</code> |
| category       | text        | no       | App category if known                         |
| created_at     | timestamp   | yes      | UTC                                           |
| updated_at     | timestamp   | yes      | UTC                                           |

## source\_apps

Maps an internal app to a source-specific app listing.

| Column            | Type       | Required | Notes                                                 |
|-------------------|------------|----------|-------------------------------------------------------|
| source_app_id     | integer pk | yes      | Internal key                                          |
| source_id         | integer fk | yes      | References <code>sources.source_id</code>             |
| app_id            | integer fk | yes      | References <code>apps.app_id</code>                   |
| external_app_id   | text       | yes      | Apple app id, e.g. <code>1058959277</code>            |
| external_app_name | text       | yes      | Apple listing name                                    |
| store_country     | text       | no       | Country used for app resolution, e.g. <code>us</code> |
| app_url           | text       | no       | App Store URL                                         |
| is_active         | boolean    | yes      | Soft-disable source app                               |
| created_at        | timestamp  | yes      | UTC                                                   |
| updated_at        | timestamp  | yes      | UTC                                                   |

Unique constraint:

- (source\_id, external\_app\_id)

## ingestion\_runs

Tracks every collector execution.

| Column                    | Type       | Required | Notes                                          |
|---------------------------|------------|----------|------------------------------------------------|
| ingestion_run_id          | text pk    | yes      | UUID or timestamp id                           |
| source_id                 | integer fk | yes      | References <code>sources.source_id</code>      |
| collector_name            | text       | yes      | Example: <code>apple_review_gui</code>         |
| collector_version         | text       | no       | Git SHA or semantic version                    |
| started_at                | timestamp  | yes      | UTC                                            |
| finished_at               | timestamp  | no       | UTC                                            |
| status                    | text       | yes      | <code>running, success, partial, failed</code> |
| requested_apps            | json/text  | no       | App keys or ids requested                      |
| requested_countries       | json/text  | no       | Country codes requested                        |
| max_pages_per_country     | integer    | no       | Collector config                               |
| delay_seconds             | real       | no       | Collector config                               |
| records_seen              | integer    | yes      | Raw candidate count                            |
| records_inserted          | integer    | yes      | New rows                                       |
| records_updated           | integer    | yes      | Existing rows updated                          |
| records_skipped_malformed | integer    | yes      | Parse failures                                 |
| records_skipped_duplicate | integer    | yes      | Dedupe skips                                   |
| error_message             | text       | no       | Failure context                                |
| created_at                | timestamp  | yes      | UTC                                            |

## reviews

One source review. This is the core fact table.

| Column           | Type       | Required | Notes                                                   |
|------------------|------------|----------|---------------------------------------------------------|
| review_pk        | integer pk | yes      | Internal key                                            |
| source_app_id    | integer fk | yes      | References <code>source_apps.source_app_id</code>       |
| ingestion_run_id | text fk    | yes      | References <code>ingestion_runs.ingestion_run_id</code> |
| source_review_id | text       | yes      | Apple review id                                         |

| Column              | Type      | Required | Notes                                     |
|---------------------|-----------|----------|-------------------------------------------|
| country             | text      | yes      | Review feed country, e.g. <code>us</code> |
| language_code       | text      | no       | Detected or source-provided ISO-like code |
| language_confidence | real      | no       | Later language detection confidence       |
| author_name         | text      | no       | Raw author display name                   |
| rating              | integer   | yes      | Expected 1-5                              |
| app_version         | text      | no       | Version reported by Apple feed            |
| title               | text      | no       | Raw title                                 |
| review_text         | text      | yes      | Raw review body                           |
| review_text_hash    | text      | yes      | SHA-256 of normalized review text         |
| review_date         | timestamp | yes      | Source review timestamp                   |
| source_updated_at   | timestamp | no       | Source updated timestamp if different     |
| collected_at        | timestamp | yes      | When collector saw this row               |
| first_seen_at       | timestamp | yes      | First time stored                         |
| last_seen_at        | timestamp | yes      | Last time observed                        |
| raw_payload         | json/text | no       | Optional raw source entry                 |
| created_at          | timestamp | yes      | UTC                                       |
| updated_at          | timestamp | yes      | UTC                                       |

Unique constraint:

- (`source_app_id`, `source_review_id`)

Recommended indexes:

- (`source_app_id`, `country`, `review_date`)
- (`rating`)
- (`app_version`)
- (`review_text_hash`)
- (`language_code`)

## review\_quality

Derived quality and deduplication signals. Keep this separate from raw review data so quality logic can evolve.

| Column       | Type          | Required | Notes                                     |
|--------------|---------------|----------|-------------------------------------------|
| review_pk    | integer pk/fk | yes      | References <code>reviews.review_pk</code> |
| title_length | integer       | yes      | Character count                           |

| Column                               | Type      | Required | Notes                            |
|--------------------------------------|-----------|----------|----------------------------------|
| <code>review_text_length</code>      | integer   | yes      | Character count                  |
| <code>review_word_count</code>       | integer   | yes      | Token-ish word count             |
| <code>ascii_ratio</code>             | real      | yes      | Useful language/noise proxy      |
| <code>has_emoji</code>               | boolean   | yes      | Emoji signal                     |
| <code>is_low_signal</code>           | boolean   | yes      | Short/generic/repeated text      |
| <code>is_duplicate_text</code>       | boolean   | yes      | Duplicate normalized text        |
| <code>duplicate_group_key</code>     | text      | no       | Hash/key for duplicate clusters  |
| <code>missing_required_fields</code> | json/text | no       | Missingness diagnostics          |
| <code>quality_flags</code>           | json/text | no       | Flexible list of flags           |
| <code>computed_at</code>             | timestamp | yes      | UTC                              |
| <code>quality_version</code>         | text      | yes      | Example: <code>quality_v1</code> |

## `review_nlp_annotations`

Future extension table for sentiment analysis and NLP outputs. Multiple rows per review are allowed so model versions can coexist.

| Column                       | Type       | Required | Notes                                                                    |
|------------------------------|------------|----------|--------------------------------------------------------------------------|
| <code>annotation_id</code>   | integer pk | yes      | Internal key                                                             |
| <code>review_pk</code>       | integer fk | yes      | References <code>reviews.review_pk</code>                                |
| <code>task_name</code>       | text       | yes      | <code>sentiment</code> , <code>topic</code> , <code>intent</code> , etc. |
| <code>model_name</code>      | text       | yes      | Model/provider identifier                                                |
| <code>model_version</code>   | text       | no       | Version or checkpoint                                                    |
| <code>label</code>           | text       | no       | Main class label                                                         |
| <code>score</code>           | real       | no       | Confidence or scalar result                                              |
| <code>scores_json</code>     | json/text  | no       | Full score map                                                           |
| <code>annotation_json</code> | json/text  | no       | Structured task output                                                   |
| <code>created_at</code>      | timestamp  | yes      | UTC                                                                      |

Unique constraint:

- `(review_pk, task_name, model_name, model_version)`

## CSV-to-v1 Mapping

| Current CSV Field | v1 Destination                                                                                       |
|-------------------|------------------------------------------------------------------------------------------------------|
| source            | <code>sources.source_key</code>                                                                      |
| app_id            | <code>source_apps.external_app_id</code>                                                             |
| app_name          | <code>source_apps.external_app_name</code> ; also maps to <code>apps.canonical_name</code>           |
| app_key           | <code>apps.app_key</code>                                                                            |
| country           | <code>reviews.country</code>                                                                         |
| review_id         | <code>reviews.source_review_id</code>                                                                |
| author            | <code>reviews.author_name</code>                                                                     |
| rating            | <code>reviews.rating</code>                                                                          |
| version           | <code>reviews.app_version</code>                                                                     |
| title             | <code>reviews.title</code>                                                                           |
| review_text       | <code>reviews.review_text</code>                                                                     |
| review_date       | <code>reviews.review_date</code>                                                                     |
| collected_at      | <code>reviews.collected_at</code> ; initial <code>first_seen_at</code> and <code>last_seen_at</code> |

## Ingestion Behavior

For every ingestion run:

1. Create an `ingestion_runs` row with `status='running'`.
2. Resolve or create `sources`, `apps`, and `source_apps`.
3. Parse each raw review entry.
4. Skip malformed rows missing source review id, rating, review text, or review date.
5. Compute `review_text_hash` from normalized text.
6. Upsert into `reviews` by (`source_app_id`, `source_review_id`).
7. On insert, set `first_seen_at=collected_at` and `last_seen_at=collected_at`.
8. On repeat observation, update `last_seen_at`, mutable metadata, and `updated_at`.
9. Compute or refresh `review_quality`.
10. Finish `ingestion_runs` with record counts and status.

## Open Decisions For v2

- Whether to store raw Apple JSON payload for every review or only for debug samples.
- Whether language detection should run during ingestion or as a downstream batch job.
- Whether duplicate text should be scoped per app, per country, or globally.
- Whether app metadata history needs slowly changing dimensions.
- Whether to move JSON/text columns to native JSONB in PostgreSQL.